



Advanced Time Series Analysis and Forecasting

Chapter 3: Structural VAR and local projections



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Acronyms used in this material (1/2)

- ▣ AB: the AB model of a structural VAR (A times residuals = B times shocks)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ ARDL: AutoRegressive Distributed Lag (model)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BQ: Blanchard–Quah (long-run identification)
- ▣ CEE: Christiano–Eichenbaum–Evans (recursive monetary VAR, 1999)
- ▣ CPI: Consumer Price Index
- ▣ CPIAUCSL: Consumer Price Index for All Urban Consumers (FRED series code)
- ▣ DGP: Data-Generating Process
- ▣ EA: Euro Area
- ▣ ECB: European Central Bank
- ▣ EIA: US Energy Information Administration
- ▣ FEVD: Forecast Error Variance Decomposition
- ▣ FF4: surprise in the federal funds futures contract three months ahead
- ▣ FOMC: Federal Open Market Committee
- ▣ FRA: Forward Rate Agreement
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GDP: Gross Domestic Product
- ▣ GK: Gertler–Karadi (proxy SVAR, 2015)
- ▣ GNP: Gross National Product
- ▣ GNPC96: Real Gross National Product (FRED series code)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HICP: Harmonised Index of Consumer Prices

Acronyms used in this material (2/2)

- HQ: Hannan–Quinn (information criterion)
- IGBREA: Index of Global Real Economic Activity (FRED series code)
- IP: Industrial Production
- JK: Jarociński–Karadi (monetary policy and information shocks, 2020)
- LLM: Large Language Model
- LNS14000025: Unemployment rate, men aged 20 and over (FRED series code)
- LP: Local Projection
- LP-IV: Local Projection with Instrumental Variables
- MA: Moving Average
- MSE: Mean Squared Error
- OLS: Ordinary Least Squares
- RMSE: Root Mean Squared Error
- ROBOR: Romanian Interbank Offer Rate
- RZ: Ramey–Zubairy (state-dependent multipliers, 2018)
- SCA: Seasonally and Calendar Adjusted
- SE: Standard Error
- SVAR: Structural Vector AutoRegression
- TSA: Time Series Analysis
- US: United States
- VAR: Vector AutoRegression
- VECM: Vector Error Correction Model

Today's question and route

- **Question:** what does a monetary policy shock, an oil supply shock or a fiscal shock *cause*, and how sure can we be?
 - ▶ a reduced-form VAR describes correlations; a causal answer needs an identifying assumption, stated and defended
- **Route** of the chapter
 - ▶ the identification problem; short-run (recursive and non-recursive) and long-run restrictions
 - ▶ sign, narrative and heteroskedasticity-based identification; set identification
 - ▶ external instruments and high-frequency surprises; inference for impulse responses
 - ▶ local projections: LP against VAR, LP-IV, state dependence; Romania and euro-area spillovers
- We build on [TSA, Chapter 6](#) (reduced-form VAR, Granger causality, Cholesky responses, FEVD) and on [Chapter 0](#) (HAC, bootstrap); Seminar 3 comes before this lecture

Learning outcomes

- ▣ State the identification problem of a structural VAR, count the restrictions it needs and explain observational equivalence through rotations
- ▣ Identify shocks by short-run, long-run, sign, narrative and heteroskedasticity restrictions, and say what each assumes about the economy
- ▣ Estimate a proxy SVAR with an external instrument, check its relevance and use inference that stays valid with weak instruments
- ▣ Build bootstrap bands for impulse responses (residual, wild, block; Kilian bias correction) and read them correctly
- ▣ Estimate local projections, LP-IV and state-dependent LP with HAC or lag-augmented inference, and choose between LP and VAR by the bias–variance trade-off

Reading, data and tools

- Backbone: Kilian and Lütkepohl (2017), Ch. 8–15 (identification, inference, LP); Hamilton (1994), Ch. 11; Lütkepohl (2005)
 - ▶ surveys: Ramey (2016) (shocks and their propagation), Stock and Watson (2018) (external instruments), Stock and Watson (2001)
- Python Quantlets of this chapter: [Quantlets/Ch_03](#)
 - ▶ VAR, Cholesky, long-run and sign identification, proxy SVAR, bootstraps, LP, LP-IV written out in `numpy`
- Lecture notebook: [open in Google Colab](#)

Data used in this chapter

Series	Source	Use
US monthly macro data, FF4 surprises, Romer–Romer shocks	replication files of Ramey (2016) (public)	CEE, sign, proxy, LP
world crude oil production; refiner acquisition cost	EIA (bulk file, Monthly Energy Review)	Kilian
global real activity index; US CPI; real GNP; unemployment of men 20+	FRED (IGREA, CPIAUCSL, GNPC96, LNS14000025)	Kilian, BQ
US quarterly fiscal data 1889–2015	replication files of Ramey and Zubairy (2018)	state-dependent LP
IP, HICP, 3-month rates (RO, EA); EUR/RON	Eurostat; BNR reference rate	Romanian VAR
ECB monetary policy shocks; EA 1-year rate	Jarociński and Karadi (2020) (authors' update); ECB Data Portal	spillovers

- All sources are public and need no account or key; monthly data up to mid-2026, except the replication files
- IP: industrial production; HICP: harmonised index of consumer prices; FF4: surprise in the fourth federal funds futures contract

From correlations to shocks



Christopher Sims, Nobel lecture, 2011

Photo: Holger Motzkau (2011); CC BY-SA 3.0; Wikimedia

[Commons](#)

- 1980: Sims (1980) proposes VARs instead of large models with “incredible” exclusion restrictions
 - ▶ the Cholesky ordering was his first identification scheme, and the first target of criticism
- 1989: long-run restrictions Blanchard and Quah (1989); 1999: the recursive monetary VAR Christiano, Eichenbaum and Evans (1999)
- 2005: sign restrictions Uhlig (2005) and local projections Jordà (2005); 2012–2018: external instruments Stock and Watson (2012), Mertens and Ravn (2013), Gertler and Karadi (2015)
- 2011: Sims and Sargent share the Nobel Prize “for their empirical research on cause and effect in the macroeconomy”
- The question has stayed the same; the identifying assumptions have become weaker and more transparent

Reduced form and structural form

- Reduced form (TSA, Chapter 6): $y_t = \nu + A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t$, $\mathbb{E} u_t u_t' = \Sigma_u$
 - ▶ u_t are one-step forecast errors: they mix all the economic shocks that hit in period t
- Structural form: $u_t = B_0 \varepsilon_t$, with structural shocks ε_t mutually uncorrelated, $\mathbb{E} \varepsilon_t \varepsilon_t' = I_n$
 - ▶ equivalently $B_0^{-1} y_t = B_0^{-1} \nu + B_0^{-1} A_1 y_{t-1} + \dots + \varepsilon_t$: contemporaneous relations among the variables
 - ▶ a **shock** is primitive, exogenous, uncorrelated with other shocks and economically meaningful [Ramey \(2016\)](#)
- Structural impulse responses: $\Theta_h = \Phi_h B_0$, with Φ_h the reduced-form MA coefficients
 - ▶ element (i, j) of Θ_h : effect of a one-standard-deviation shock j on variable i after h periods

Counting: the data do not pin down B_0

- The data identify $\Sigma_u = B_0 B_0'$: $n(n+1)/2$ distinct moments; B_0 has n^2 unknowns
 - ▶ order condition: at least $n(n-1)/2$ restrictions (3 for $n=3$, 28 for $n=8$)
 - ▶ the rank condition must also hold locally [Rubio-Ramírez, Waggoner and Zha \(2010\)](#): counting is necessary, not sufficient
- **Observational equivalence**: for any orthogonal Q ($QQ' = I$), $\tilde{B}_0 = B_0 Q$ gives $\tilde{B}_0 \tilde{B}_0' = \Sigma_u$
 - ▶ the two models fit the data equally well and imply different impulse responses
 - ▶ identification = a rule that selects one Q (point identification) or a set of Q (set identification)
- A useful parametrisation: $B_0 = PQ$ with P the Cholesky factor of Σ_u ; every identification scheme is a choice of Q

Invertibility and the limits of a VAR

- A VAR recovers ε_t from current and past y_t only if the shocks are **invertible** (fundamental)
 - ▶ news shocks and fiscal foresight break invertibility: agents react before the econometrician sees the shock
 - ▶ remedy: add forward-looking variables (asset prices, forecasts) or use the shock directly [Ramey \(2016\)](#)
- With an external instrument we need only *partial* invertibility of the shock of interest [Stock and Watson \(2018\)](#)
 - ▶ local projections with the shock as regressor do not need invertibility at all [Plagborg-Møller and Wolf \(2021\)](#)
- Practical rule: a small VAR omitting information the agents use (expectations, commodity prices) produces “puzzles”

Recap: The identification problem

- ▣ Σ_u fixes B_0 only up to an orthogonal rotation Q
- ▣ Point identification needs at least $n(n-1)/2$ restrictions and the rank condition
- ▣ Every scheme is an economic assumption: state it, defend it, test what can be tested
- ▣ Invertibility is a separate assumption; instruments and LP relax it

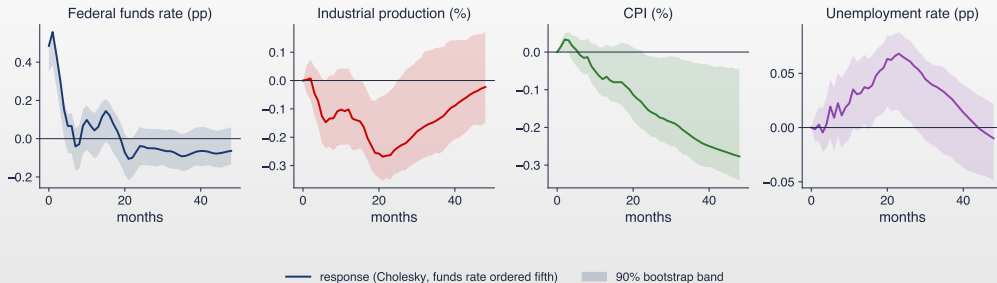
Recursive identification

- $B_0 = P$ lower triangular: variable i does not respond within the period to shocks ordered after it
 - ▶ exactly $n(n-1)/2$ zeros: just identified; the ordering is the assumption (TSA, Chapter 6 showed that it matters)
- Monetary block of Christiano, Eichenbaum and Evans (1999): slow variables (output, prices), the policy rate, fast variables (reserves, money)
 - ▶ assumption: within a month, output and prices do not react to policy, and policy reacts to them
 - ▶ only the position of the policy variable matters for its shock; the ordering inside the blocks is irrelevant
- Plausible at a monthly frequency with slow-moving real variables; implausible for asset prices or for quarterly data

Case study: Christiano, Eichenbaum and Evans (1999)

- The paper: the benchmark recursive monetary VAR; US monthly data, the federal funds rate shock ordered after output, prices and commodity prices [Christiano, Eichenbaum and Evans \(1999\)](#)
 - ▶ we use the specification of [Ramey \(2016\)](#), Figure 1: VAR(12) in log IP, unemployment, log CPI, log commodity prices, funds rate, log nonborrowed and total reserves, log M1
- Sample 1965:1–1995:6 ($T = 354$), constant, Cholesky with the funds rate fifth; 90% bands from 500 residual-bootstrap replications
 - ▶ the largest root of the companion matrix is 0.9999: a VAR in levels, consistent for responses at short and medium horizons [Kilian and Lütkepohl \(2017\)](#)
- Question: how do output and prices respond to an unexpected tightening?

A recursive monetary policy shock (CEE specification)



☐ Responses to a one-standard-deviation funds-rate shock (0.48 pp); 90% residual-bootstrap bands

ATS_ch3_recursive_var

Interpreting the CEE responses

- Industrial production falls slowly: trough -0.27% after 21 months (band $[-0.35, -0.05]$); unemployment rises by up to 0.07 pp
 - ▶ the hump shape and the delay are the classic “long and variable lags” of monetary policy
- Prices first rise slightly (0.033% after 2 months), then fall to -0.28% after four years
 - ▶ the initial rise is the **price puzzle**: the Fed tightens when it expects inflation; commodity prices in the VAR reduce but do not remove it
- Monetary shocks explain little of output: 6.6% of the 24-month forecast error variance of IP

Non-recursive short-run restrictions

- **AB model** Lütkepohl (2005), Kilian and Lütkepohl (2017): $Au_t = B\varepsilon_t$; zeros and known values anywhere in A and B
 - ▶ estimated by maximum likelihood: maximise $-\frac{T}{2} \ln |A^{-1}BB'A^{-1}| - \frac{T}{2} \text{tr}[(A^{-1}BB'A^{-1})^{-1}\hat{\Sigma}_u]$
 - ▶ over-identifying restrictions can be tested by a likelihood-ratio test against the just-identified model
- **Blanchard–Perotti** Blanchard and Perotti (2002): the automatic response of taxes to output is an *external* elasticity (from tax codes)
 - ▶ with that elasticity fixed, the cyclically adjusted tax innovation is an instrument for the remaining coefficients
 - ▶ government spending is predetermined within the quarter (decision lags): a zero restriction with an institutional argument
- Seminar 3, A2: one known elasticity identifies a 2×2 system by instrumental variables

What the recursive assumption cannot do

- Simultaneity: interest rates and asset prices react to each other within the day
 - ▶ no ordering of stock prices and the policy rate is credible at a monthly frequency
- Omitted information: the policy shock absorbs the Fed's forecasts
 - ▶ narrative shocks [Romer and Romer \(2004\)](#) purge the intended rate change of the Greenbook forecasts
- Sample dependence: the post-1983 responses are weaker and often puzzling [Ramey \(2016\)](#)
 - ▶ the remedies of the next sections: weaker restrictions (signs) or outside information (instruments)

Recap: Short-run restrictions

- Cholesky = timing assumptions; only the position of the shock variable matters for its responses
- CEE: output falls with a lag of about 21 months; the price puzzle remains
- Non-recursive schemes use institutional knowledge (tax elasticities, decision lags) and can be over-identified and tested
- Short-run zeros are least credible exactly where they matter most: fast markets

Not all oil price shocks are alike

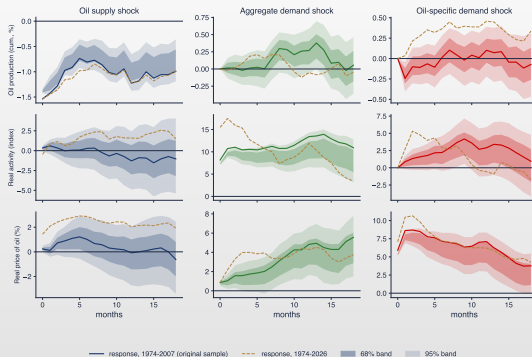


Burning Kuwaiti oil field, March 1991

Photo: JO1 Gawlowicz, US Navy (1991); public domain;
Wikimedia Commons

- Kilian (2009): monthly VAR(24) in (i) the percent change of world crude oil production, (ii) global real activity, (iii) the log real price of oil
 - ▶ recursive: oil supply does not respond within the month to demand (adjustment costs); real activity does not respond within the month to oil-specific demand
- Three shocks: oil supply, aggregate demand, oil-specific (precautionary) demand
 - ▶ real activity: the index of Kilian (2009) as corrected in Kilian (2019) (FRED IGREA); price: refiner acquisition cost of imported crude, deflated by the US CPI
- Our sample starts in 1974 (first month of the price series), effective sample from January 1976, $T = 384$; extension to May 2026

Responses to oil supply and demand shocks

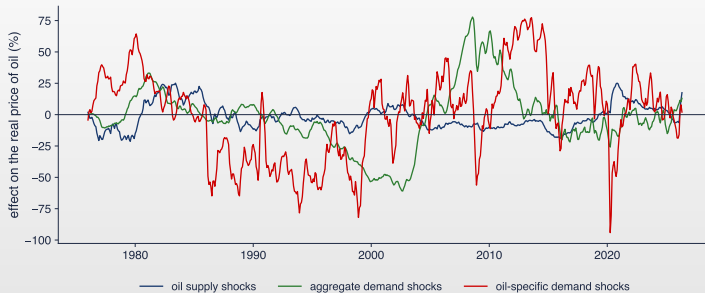


- One-standard-deviation shocks, production cumulated; 68% and 95% recursive-design wild-bootstrap bands [Gonçalves and Kilian \(2004\)](#); dashed: sample extended to 2026

Interpreting the oil responses

- ▣ Supply shock: production falls by -1.5% on impact, the real price barely moves (-0.1% after 12 months, band $[-2.7, 2.1]$)
 - ▶ in the extended sample the effect is larger (2.0%): the post-2008 market reacts more to supply news
- ▣ Aggregate demand: a delayed, persistent rise of the price (4.8% after 12 months, band $[1.6, 6.7]$)
- ▣ Oil-specific demand: an immediate jump (5.9%) that persists (7.1% after 12 months)
- ▣ At 60 months, supply shocks explain 1% of the price variance, aggregate demand 64% , oil-specific demand 35% : oil prices are mostly demand-driven

Historical decomposition of the real price of oil



▣ Cumulative contribution of each structural shock, 1976–2026 (recursive VAR(24), extended sample); log points $\times 100$

Q ATS_ch3_oil_shocks

Interpreting the historical decomposition

- ▣ 2003–mid-2008: aggregate demand adds 126 log points, supply -12 : the boom was a global demand boom
- ▣ Mid-2008 to early 2009: oil-specific demand -82 , aggregate demand -35 ; early 2020: oil-specific demand -103
- ▣ Mid-2021 to mid-2022: oil-specific demand 43, supply -2 : the model reads the 2022 surge as precautionary demand
- ▣ Caveat: the zero restriction on the supply response is strong; [Baumeister and Hamilton \(2019\)](#) allow a small but non-zero elasticity and find a larger role for supply
 - ▶ a decomposition is only as good as the identifying assumption behind it

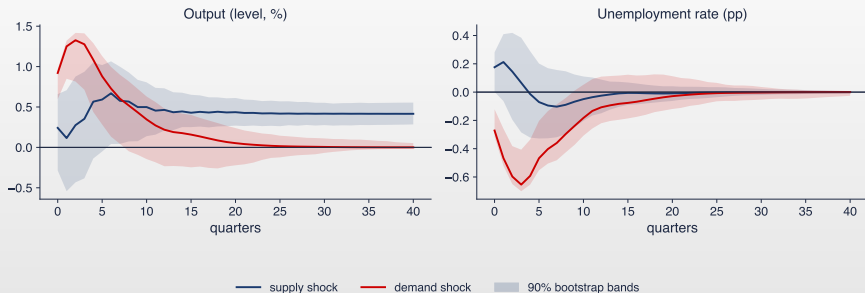
Blanchard and Quah: the long-run multiplier

- For $y_t = (\Delta \text{output}_t, u_t)'$, the cumulative response of output is $\Theta(1) = \sum_h \Theta_h = (I - A_1 - \dots - A_p)^{-1} B_0 = A(1)^{-1} B_0$
 - ▶ restriction [Blanchard and Quah \(1989\)](#): demand shocks have no long-run effect on the level of output, $[\Theta(1)]_{12} = 0$
- Algorithm: $\Theta(1)\Theta(1)' = A(1)^{-1}\Sigma_u A(1)^{-1'}$; take its Cholesky factor L , then $B_0 = A(1)L$
 - ▶ the zero is placed on the long-run matrix, not on the impact matrix; derivation in the [Appendix](#) and in Seminar 3, A3
- Interpretation: “supply” = all shocks with permanent effects (technology, labour supply); [Galí \(1999\)](#) uses the same idea for technology shocks and hours

Case study: Blanchard and Quah (1989)

- The paper: US real GNP growth and the unemployment rate (men aged 20 and over), quarterly, VAR(8), 1950:2–1987:4 [Blanchard and Quah \(1989\)](#)
 - ▶ data treatment: separate means of output growth before and after 1973:4 (the productivity slowdown), a linear trend removed from unemployment
- Our replication: the same specification on today's vintage (FRED GNPC96, LNS14000025), $T = 151$; 90% residual-bootstrap bands
- Question: do demand shocks drive the business cycle, and how long do their effects last?

Supply and demand shocks identified by a long-run restriction



□ Output level (cumulated growth) and unemployment; one-standard-deviation shocks; 90% bootstrap bands

ATS_ch3_long_run

Interpreting the Blanchard–Quah responses

- Demand shock: output peaks at 1.33% after 2 quarters and returns to zero by construction; unemployment falls by -0.65 pp
- Supply shock: output rises permanently (0.42% after ten years, band $[0.28, 0.55]$), unemployment first *rises* (0.18 pp)
- Demand shocks explain 92% of the output-level variance at one year and 54% at ten years: as in the paper, demand dominates at business-cycle horizons
- Seminar 3, B5: without the break and the trend, the shares change dramatically

How reliable are long-run restrictions?

- Faust and Leeper (1997): the long-run multiplier $A(1)^{-1}$ is estimated imprecisely from finite samples
 - ▶ a shock with a tiny but permanent effect is indistinguishable from a transitory one: inference can be arbitrarily distorted
 - ▶ the near-unit-root behaviour of unemployment makes the problem worse
- Results depend on the treatment of low frequencies: breaks in mean growth, trends, the lag length
 - ▶ Chapter 2 tests for such breaks; Chapter 4 treats cointegration, where long-run restrictions become restrictions on the VECM
- Use long-run restrictions with a sensitivity table, never alone

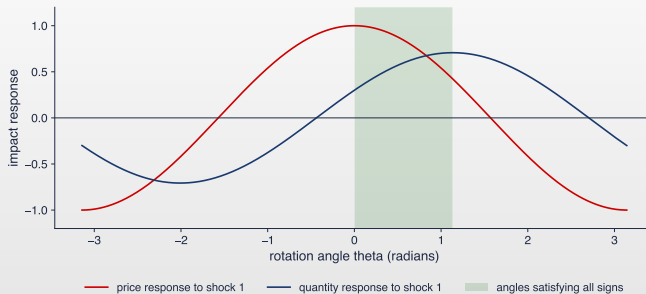
Recap: Long-run restrictions

- ▣ Restrict $\Theta(1) = A(1)^{-1}B_0$ instead of B_0 : identification from permanence
- ▣ BQ: demand dominates business-cycle fluctuations in output, supply the long run
- ▣ Fragile: low-frequency treatment and the imprecise long-run multiplier drive the answer

Identification by signs

- ▣ Restrict only the **signs** of some responses for some horizons: a contractionary monetary shock raises the rate and lowers prices Uhlig (2005)
 - ▶ the response of interest (output) is left unrestricted: an “agnostic” procedure
- ▣ Algorithm Rubio-Ramírez, Waggoner and Zha (2010): draw X with i.i.d. $N(0, 1)$ entries, $X = QR$, normalise the signs of $\text{diag}(R)$; Q is Haar-uniform on the orthogonal group
 - ▶ compute $\Theta_h = \Phi_h P Q$ and keep Q if all signs hold; repeat for draws of (A, Σ_u) from the posterior
 - ▶ zero restrictions together with signs need the importance-sampling algorithm of Arias, Rubio-Ramírez and Waggoner (2018)
- ▣ The result is a **set** of models consistent with the data and the restrictions, not a point

Sign restrictions in two dimensions



- Price and quantity innovations with variances 1 and 0.5 and covariance 0.3; impact responses $P(\cos \theta, \sin \theta)'$ to shock 1 as the rotation angle varies; demand: (+, +), supply: (+, -)

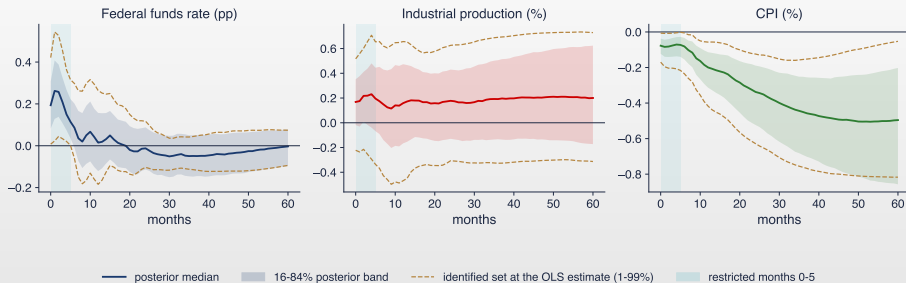
Interpreting the rotation set

- ▣ All signs hold for $\theta \in (0, 1.13)$, 18% of the circle: a whole interval of models is equally consistent with the data
- ▣ The impact effect of demand on quantity lies anywhere in $[0.30, 0.71]$: the **identified set**
- ▣ A uniform draw of θ puts a prior on this interval: the posterior inside the set is the prior, not information from data
 - ▶ this is the point of [Baumeister and Hamilton \(2015\)](#): Haar draws are informative about the responses

Case study: Uhlig (2005)

- The paper: six-variable monthly US VAR(12), no constant, 1965:1–2003:12; a contractionary shock raises the funds rate and lowers prices, commodity prices and nonborrowed reserves for months 0–5 [Uhlig \(2005\)](#)
 - ▶ output is not restricted: the question is its sign
- Our replication: the same restrictions and sample; IP and CPI replace the interpolated monthly GDP and deflator of the paper (data of [Ramey \(2016\)](#))
 - ▶ flat Normal–inverse-Wishart posterior, 1 000 draws, Haar impulse vectors; accepted: 985 draws; at the OLS estimate 4.1% of rotations satisfy the signs
- Question: does a contractionary monetary shock lower output when output is left free?

An agnostic monetary policy shock



- One-standard-deviation shock; posterior median and 16–84% band; dashed: 1–99% range of the identified set at the OLS estimate; shaded: restricted months

Interpreting the agnostic responses

- ▣ Output after 12 months: median 0.16%, band $[-0.17, 0.47]$; 32% of draws negative: no evidence that output falls
- ▣ The identified set at the OLS estimate spans $[-0.47, 0.66]$: the data with the signs cannot sign output
- ▣ Prices fall persistently (-0.34% after 24 months) without a price puzzle, because the restriction rules it out
- ▣ Uhlig's conclusion, confirmed: neutrality of monetary policy shocks for output is not inconsistent with the data

Inference under set identification

- The posterior median is not a structural model: at each horizon it can come from a different rotation [Fry and Pagan \(2011\)](#)
 - ▶ report a single admissible model close to the median (median target) or, better, the whole set
- The Haar prior is informative about the responses [Baumeister and Hamilton \(2015\)](#); [Giacomini and Kitagawa \(2021\)](#) propose robust Bayes: report the range of posterior means over all priors on Q
 - ▶ the robust band is wider and honest: it separates what the data say from what the prior says
- Tighten the set with more information: more signs, zeros [Arias, Rubio-Ramírez and Waggoner \(2018\)](#), narrative events, elasticity bounds [Baumeister and Hamilton \(2019\)](#)

Narrative restrictions



Paul Volcker, Fed Chairman 1979–1987

Photo: Federal Reserve (2014); public domain;
Wikimedia Commons

- Antolín-Díaz and Rubio-Ramírez (2018): restrict the structural shocks at specific dates, not the responses
 - ▶ October 1979 (the Volcker shift to reserve targeting) was a positive monetary policy shock
 - ▶ and it was the main contributor to the unexpected change of the funds rate that month
- A few credible events remove most of the rotations that sign restrictions alone leave
 - ▶ the likelihood must be reweighted by the probability of satisfying the narrative (importance sampling)
- With narrative restrictions output falls after a contractionary shock, unlike in the agnostic set

Identification through heteroskedasticity

- If the variances of the structural shocks change across regimes while B_0 stays fixed: $\Sigma_1 = B_0 B_0'$, $\Sigma_2 = B_0 \Lambda B_0'$, Λ diagonal [Rigobon \(2003\)](#)
 - ▶ then $\Sigma_1^{-1} \Sigma_2 = B_0^{-1'} \Lambda B_0'$: the columns of B_0 follow from an eigendecomposition, unique if the λ_i are distinct
 - ▶ regimes: known dates (crises, policy announcements) or estimated (Markov switching, GARCH) [Lanne and Lütkepohl \(2008\)](#)
- Statistical identification gives shocks without names: labelling them still needs economics (signs)
 - ▶ the assumption that B_0 is the same in both regimes is strong and testable only partly
- Seminar 3, A8 recovers B_0 from two covariance matrices by hand; [Chapter 7](#) develops regime switching

Recap: Sign restrictions and set identification

- ▣ Signs give a set of models; report the set, not only the posterior median
- ▣ Uhlig: with prices, rates and reserves restricted, output can go either way
- ▣ The Haar prior matters; robust Bayes and narrative restrictions sharpen honest conclusions
- ▣ Heteroskedasticity identifies statistically; economics still names the shocks

The proxy SVAR

- An external variable z_t (the proxy) with **relevance** $\mathbb{E}z_t\varepsilon_{1t} = \alpha \neq 0$ and **exogeneity** $\mathbb{E}z_t\varepsilon_{jt} = 0, j \neq 1$
 - ▶ then $\mathbb{E}u_t z_t = B_0 \mathbb{E}\varepsilon_t z_t = \alpha b_1$: the covariance of the residuals with the proxy is proportional to the impact column b_1
- Normalise $b_{11} = 1$ (or a 25 bp rate increase): $b_{i1}/b_{11} = \mathbb{E}(u_{it}z_t)/\mathbb{E}(u_{1t}z_t)$, a 2SLS of u_{it} on u_{1t} with instrument z_t [Stock and Watson \(2012\)](#), [Mertens and Ravn \(2013\)](#)
 - ▶ responses $\Theta_{h, \cdot 1} = \Phi_h b_1$; the proxy may be noisy, short or available only for a sub-sample
- Only one column of B_0 is identified, and no zero restriction is needed

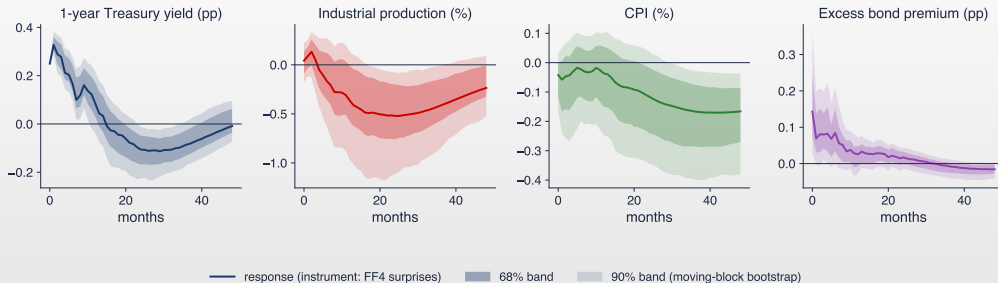
High-frequency identification of monetary policy

- Surprise = change of futures-implied rates in a narrow window (30 minutes) around an FOMC announcement [Kuttner \(2001\)](#), [Gürkaynak, Sack and Swanson \(2005\)](#)
 - ▶ within 30 minutes nothing else systematic happens: exogeneity by design
 - ▶ FF4: the three-month-ahead federal funds futures; it captures forward guidance as well as the current decision [Gertler and Karadi \(2015\)](#)
- Complications: the **information effect** [Nakamura and Steinsson \(2018\)](#): a tightening can signal good news about the economy
 - ▶ [Jarociński and Karadi \(2020\)](#) separate pure policy and information shocks with the co-movement of rates and stock prices (sign restrictions on surprises)
 - ▶ surprises are predictable from past macro news [Bauer and Swanson \(2023\)](#): orthogonalise them before use

Case study: Gertler and Karadi (2015)

- The paper: the 1-year Treasury yield as policy indicator (it captures forward guidance), log IP, log CPI and the excess bond premium of Gilchrist and Zakrajšek; VAR(12), 1979:7–2012:6 [Gertler and Karadi \(2015\)](#)
 - ▶ instrument: FF4 surprises, monthly, 1991:1–2012:6; the residuals of the full sample are used for the proxy regression on the overlap
- Our replication: the same variables, sample and instrument (replication files of [Ramey \(2016\)](#)), $T = 384$, 258 months with the instrument
 - ▶ first stage: robust $F = 17.7$ (homoskedastic 21.9), $R^2 = 7.8\%$; 90% moving-block bootstrap bands [Jentsch and Lunsford \(2019\)](#)
- Question: does a policy surprise move credit spreads, and with what effect on activity?

A monetary policy shock identified with FF4 surprises



□ Shock normalised to raise the 1-year yield by 25 bp on impact; 68% and 90% moving-block bootstrap bands

ATS_ch3_proxy_svar

Interpreting the proxy SVAR

- The excess bond premium jumps by 0.14 pp on impact (band $[0.03, 0.36]$): monetary policy works through credit costs, as in the paper
- IP reaches -0.52% after 25 months (band $[-1.09, -0.08]$); CPI is -0.12% after two years (band $[-0.34, 0.04]$), with no price puzzle
- $F = 17.7$ passes the rule of thumb $F > 10$, but the price response is imprecise: weak-instrument-robust bands are wider [Montiel Olea, Stock and Watson \(2021\)](#)

Inference with an external instrument

- Weak instruments: when α is small, the ratio $\mathbb{E}(u_i z)/\mathbb{E}(u_1 z)$ is badly estimated and delta-method bands undercover
 - ▶ Montiel Olea, Stock and Watson (2021): Anderson–Rubin-type confidence sets, valid for any strength; they can be unbounded
 - ▶ report the first-stage F with HAC variance; the rule $F > 10$ is a heuristic, not a test
- Bootstrap: the wild bootstrap of Mertens and Ravn (2013) is invalid for proxy SVARs; use the moving-block bootstrap of residuals and instrument jointly Jentsch and Lunsford (2019)
 - ▶ the block keeps the dependence between the squared shock and the instrument
- Seminar 3, A5 computes relative impacts and F by hand; C1 builds an Anderson–Rubin set for Romania

Recap: External instruments

- ▣ Relevance + exogeneity of a proxy identify one column of B_0 , with no zeros
- ▣ High-frequency surprises are the best proxies we have; check the information effect and predictability
- ▣ GK replicated: credit spreads jump, output falls; $F = 17.7$
- ▣ Use weak-IV-robust sets and the block bootstrap

Delta method and its limits

- ▣ $\hat{\Theta}_h = g(\hat{\beta}, \hat{\sigma})$ is a smooth function of the VAR estimates: $\sqrt{T}(\hat{\Theta}_h - \Theta_h) \rightarrow N(0, G\Sigma G')$ [Lütkepohl \(2005\)](#)
 - ▶ $G = \partial g / \partial(\beta, \sigma)$ in closed form; quick, but symmetric bands
- ▣ It fails where we need it most
 - ▶ near unit roots and at long horizons the distribution of $\hat{\Theta}_h$ is skewed and non-Normal
 - ▶ the covariance $G\Sigma G'$ can be singular (e.g. when a response is zero by construction)
 - ▶ OLS slope estimates are biased towards zero in small samples, so responses decay too fast
- ▣ Hence the bootstrap, and the bias correction

Bootstrap schemes for VARs

- ▣ **Residual (recursive-design)**: resample $\hat{u}_t$ i.i.d., rebuild y_t^* recursively from the estimated VAR, re-estimate, identify, repeat
 - ▶ valid with i.i.d. innovations; the percentile (or Hall) interval of the B responses gives the band
- ▣ **Wild**: $u_t^* = \eta_t \hat{u}_t$, $\eta_t = \pm 1$ with probability $1/2$: robust to conditional heteroskedasticity [Gonçalves and Kilian \(2004\)](#)
 - ▶ used for the oil VAR: commodity markets have volatility clustering
- ▣ **Moving block**: resample blocks of consecutive $(\hat{u}_t, z_t)$; needed whenever higher moments matter (proxy SVAR) [Jentsch and Lunsford \(2019\)](#)
 - ▶ block length of order $T^{1/4}$; [Chapter 0](#) covered the choice

Kilian's bias-corrected bootstrap

- ▣ Kilian (1998): (1) bootstrap the bias $\hat{b} = \bar{A}^* - \hat{A}$; (2) correct $\tilde{A} = \hat{A} - \hat{b}$, shrinking the correction if $\tilde{A}$ is not stationary; (3) bootstrap again from $\tilde{A}$ and correct each replicate
 - ▶ the “bootstrap after bootstrap” recentres the distribution where OLS bias is large: persistent series, short samples
- ▣ Oil VAR: mean absolute bias 0.009 per coefficient; largest root 0.994 before, 0.9999 after the (shrunk) correction
 - ▶ 12-month response of the price to oil-specific demand: 7.11% (OLS), 7.34% (corrected), band [3.4, 9.1]; to aggregate demand: 4.83% and 4.95%
- ▣ The correction makes responses more persistent; when the VAR in levels has a root above one, the rule of Kilian leaves it uncorrected (Romanian VAR, Section 9)

Pointwise and joint bands

- A pointwise 90% band covers Θ_h at each h separately; the whole path lies inside it with much lower probability
 - ▶ questions about the *shape* (“output falls for two years”) need simultaneous bands [Montiel Olea and Plagborg-Møller \(2019\)](#)
 - ▶ sup- t band: scale the pointwise standard errors by the $1 - \alpha$ quantile of $\max_h |t_h|$ from the bootstrap
- Bayesian bands (posterior quantiles) answer a different question: probability statements given the prior
 - ▶ [Chapter 5](#) builds them with Minnesota and Normal-inverse-Wishart priors; with set identification they mix prior and data (previous section)
- Always state: pointwise or joint, frequentist or Bayesian, the level (68%, 90%, 95%)

Recap: Inference for impulse responses

- ▣ Delta-method bands are symmetric and fail near unit roots
- ▣ Residual, wild and block bootstraps match i.i.d., heteroskedastic and proxy settings
- ▣ Kilian's correction removes small-sample bias; it is not applied to explosive estimates
- ▣ Shape questions need joint bands; Bayesian bands come in [Chapter 5](#)

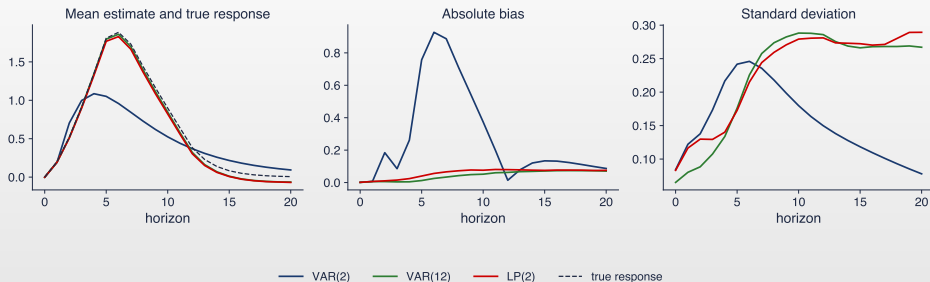
Local projections

- [Jordà \(2005\)](#): one regression per horizon, $y_{i,t+h} = \mu_h + \beta_h x_t + \gamma_h' w_t + \xi_{t+h}$, $h = 0, 1, \dots, H$
 - ▶ x_t : the shock (or the policy variable with controls w_t that make it as good as random); $\hat{\beta}_h$ is the impulse response
 - ▶ no iteration of a model: a misspecified dynamic does not compound across horizons
- The error ξ_{t+h} is serially correlated (an $MA(h-1)$ at least): HAC standard errors with $h+1$ lags, or
 - ▶ **lag augmentation** [Montiel Olea and Plagborg-Møller \(2021\)](#): add one more lag of the controls and use Eicker–Huber–White errors; valid uniformly over persistence, even near a unit root
- Flexible: nonlinear terms, state dependence, panel data, cumulative outcomes all enter as regressors

LP and VAR estimate the same responses

- Plagborg-Møller and Wolf (2021): in population, an LP with p lags of all variables as controls and a $\text{VAR}(p)$ with the same ordering give *identical* responses up to horizon p
 - ▶ with unrestricted lags ($p \rightarrow \infty$) they coincide at all horizons: the choice is about estimation, not identification
 - ▶ any VAR identification scheme (recursive, external instrument) has an LP counterpart, and vice versa
- In finite samples they differ: the VAR extrapolates the first p autocovariances, the LP uses the raw h -step covariance
 - ▶ the next slide quantifies this

Bias and variance: LP against VAR (simulation)



- DGP: $x_t = 0.5x_{t-1} + e_t$, $y_t = 0.6y_{t-1} + \sum_{j=0}^{11} \psi_j e_{t-j} + u_t$ (hump-shaped ψ_j), $T = 240$, 500 replications; a VAR(2) is misspecified

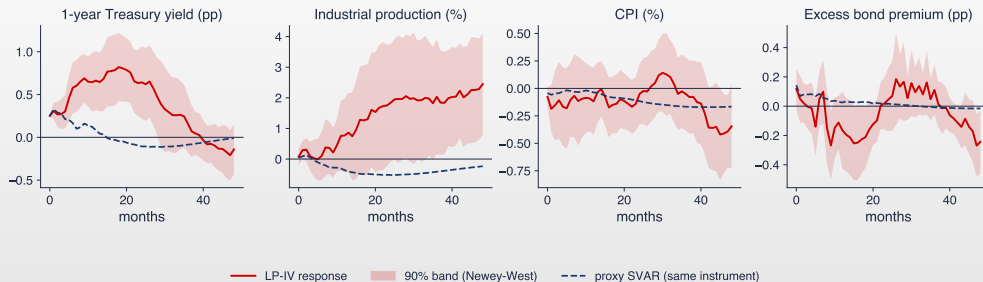
Interpreting the bias–variance trade-off

- At $h = 8$ (true response 1.44): VAR(2) bias 0.71, LP(2) bias 0.07; RMSE 0.74 and 0.27
 - ▶ where the short VAR misses the hump, its bias dominates the error
- At $h = 16$: standard deviation 0.11 (VAR(2)) against 0.27 (LP(2)); RMSE 0.17 against 0.28: the biased VAR wins
- The general lesson of [Li, Plagborg-Møller and Wolf \(2024\)](#) (thousands of DGPs): LP has low bias and high variance; VAR (and shrinkage VARs) have lower MSE unless the analyst cares almost only about bias
- A long VAR(12) behaves like LP: few lags = more bias, less variance

LP-IV

- **Stock and Watson (2018)**: $y_{i,t+h} = \theta_{h,i} y_{1t} + \gamma' w_t + \xi_{t+h}$, with y_{1t} (the policy rate) instrumented by z_t
 - ▶ $\theta_{h,i}$ = response of y_i after h periods to a shock that raises y_1 by one unit on impact
 - ▶ conditions: relevance, contemporaneous exogeneity and **lead-lag exogeneity** $\mathbb{E} z_t \varepsilon_{t+j} = 0$ for $j \neq 0$, given w_t
- Same identifying information as the proxy SVAR; different estimator
 - ▶ the proxy SVAR needs invertibility of the shock; LP-IV does not, but pays in variance
 - ▶ cumulative LP-IV gives multipliers: $\sum_{j \leq h} y_{t+j}$ on $\sum_{j \leq h} g_{t+j}$ instrumented by z_t **Ramey and Zubairy (2018)**

The same instrument, two estimators



- LP-IV with FF4 as instrument for the 1-year yield, 2 lags of all variables and of the instrument as controls, origins 1990:1–2012:6 ($T = 270$) [Ramey \(2016\)](#); dashed: the proxy SVAR

Interpreting LP-IV against the proxy SVAR

- ▣ Impact first stage $F = 21.5$; after 24 months IP: LP-IV 1.84% (SE 1.17), proxy SVAR -0.52%
- ▣ CPI after 24 months: LP-IV -0.05% (SE 0.17), proxy SVAR -0.12% : the LP bands contain the VAR path almost everywhere
- ▣ LP-IV implies a much more persistent rise of the yield: the two estimators answer slightly different questions when the policy path differs
 - ▶ the positive LP output response, noted by [Ramey \(2016\)](#), is imprecise and sensitive to the sample: do not read it as expansionary tightening

State-dependent local projections

- Interact everything with a lagged state indicator I_{t-1} :

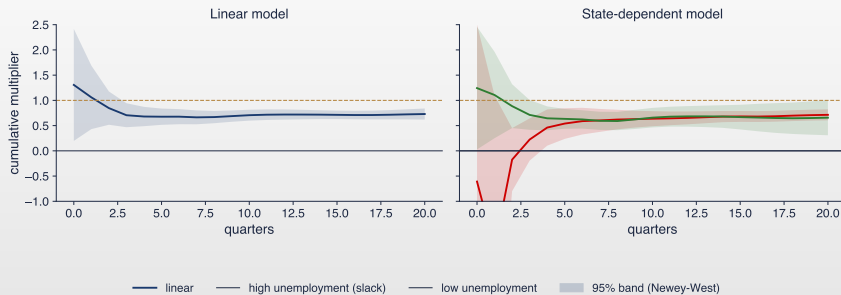
$$y_{t+h} = I_{t-1}[\alpha_{A,h} + \beta_{A,h}x_t + \dots] + (1 - I_{t-1})[\alpha_{B,h} + \beta_{B,h}x_t + \dots] + \xi_{t+h}$$

- ▶ the state may change after the shock: LP measures the average response given the initial state, including transitions
- ▶ no need to model the state dynamics, unlike a threshold or a smooth-transition VAR ([Chapter 2](#))

- Case study [Ramey and Zubairy \(2018\)](#): are government spending multipliers larger in slack times?

- ▶ US quarterly data 1889–2015; military news shock (Ramey) scaled by potential GDP; slack = unemployment $\geq 6.5\%$ in $t - 1$ (36% of quarters)
- ▶ cumulative LP-IV of output on government spending, 4 lags of news, output and spending, Newey–West with $h + 1$ lags

Government spending multipliers by state of the economy



□ Cumulative multiplier $\sum_{j \leq h} \Delta y_{t+j} / \sum_{j \leq h} \Delta g_{t+j}$ by horizon; 95% Newey–West bands; dashed: multiplier of one

Q ATS_ch3_state_dependent_lp

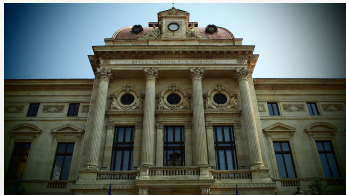
Interpreting the state-dependent multipliers

- ▣ Linear: 0.67 at two years (SE 0.06) and 0.71 at four years (SE 0.04)
- ▣ High unemployment: 0.62 and 0.68; low unemployment: 0.59 and 0.66
- ▣ As in the paper: multipliers below one in both states and no significant difference between them
- ▣ The case for LP here: a VAR would impose the same dynamics in both states unless the state transitions are modelled

Recap: Local projections

- ▣ LP = one regression per horizon; HAC or lag-augmented inference
- ▣ LP and VAR target the same responses; finite samples trade bias for variance
- ▣ LP-IV uses external instruments without invertibility, at a cost in precision
- ▣ State dependence is natural in LP: RZ find multipliers below one in slack and in normal times

Monetary policy in a small open economy



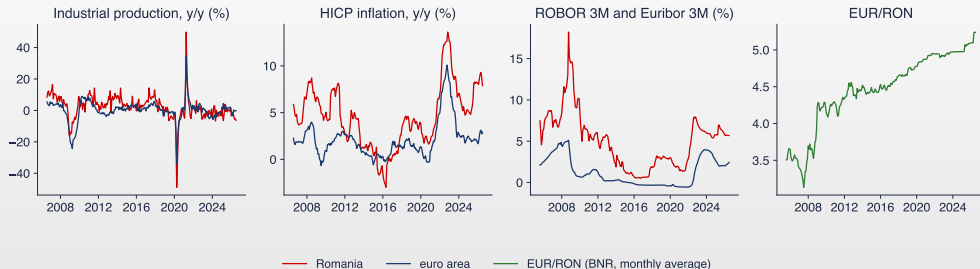
The National Bank of Romania, Bucharest

Photo: Ștefan Jurcă (2015); CC BY 2.0; Wikimedia

[Commons](#)

- The BNR targets inflation since August 2005 (target $2.5\% \pm 1$ pp since 2013) under a managed float of the leu
 - ▶ the interbank rate ROBOR 3M transmits the policy rate and liquidity conditions; we use it as the policy indicator
- Identification problems specific to Romania
 - ▶ short sample (20 years), structural change (2008–2009, 2020, 2022), administered prices and excise changes
 - ▶ external shocks dominate: the euro area takes most Romanian exports; ECB policy moves Romanian rates and the leu
- Our approach: a transparent recursive VAR with the euro area first, then an external ECB shock with LP

Romanian and euro-area data



- Monthly, August 2005–July 2026: IP (SCA), HICP, 3-month money-market rates (Eurostat), EUR/RON (BNR reference rate, monthly average)

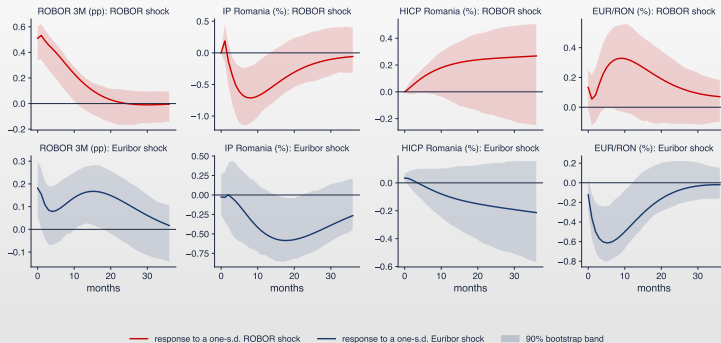
Interpreting the Romanian data

- ▣ HICP inflation peaked at 13.6% in November 2022 and is 7.9% in the last month; ROBOR 3M reached 18.2% in the 2008 liquidity squeeze
- ▣ EUR/RON rose from 3.51 to 5.24: a trend depreciation with long managed-float plateaus
- ▣ Romanian and euro-area IP co-move closely, including the 2020 collapse: a strong case for an external block
- ▣ Sharp outliers (2008, 2020, 2022) will dominate the estimates of a small VAR: check robustness without them

A VAR for Romania with a euro-area block

- $y_t = (\text{EA IP, EA HICP, Euribor 3M, RO IP, RO HICP, ROBOR 3M, EUR/RON})$, logs $\times 100$ except rates; constant and monthly dummies (HICP is not seasonally adjusted)
 - ▶ lag length: BIC and HQ choose 2, AIC 6; we use 2 ($T = 250$); largest root 1.006: levels, near-unit-root, no bias correction
- Identification (Cholesky, in this order): the euro area does not react to Romania within the month; ROBOR reacts to Romanian activity and prices within the month; the exchange rate reacts to everything
 - ▶ the zero restriction on the euro area is credible (Romania is small); the position of ROBOR before EUR/RON is debatable
 - ▶ a block-exogenous VAR would also set the lagged Romanian effects on the euro area to zero (Seminar 3, B2 discusses orderings)

Responses to a domestic and a euro-area rate shock



- VAR(2), Cholesky with the euro-area block first; one-standard-deviation shocks (ROBOR 0.51 pp, Euribor 0.09 pp); 90% residual-bootstrap bands

Interpreting the Romanian responses

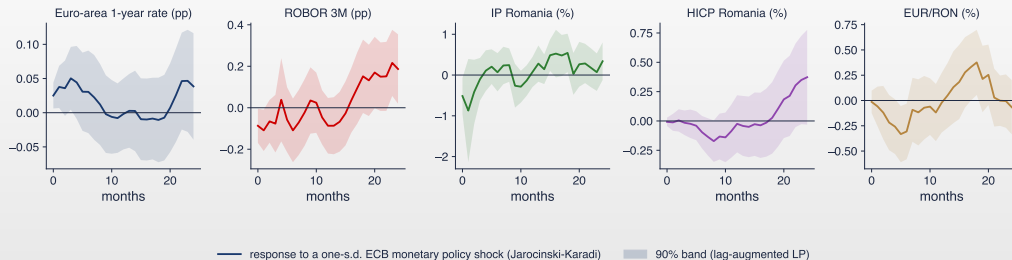
- ▣ ROBOR shock: IP -0.62% after 12 months (band $[-0.88, -0.06]$); HICP 0.20% (band $[-0.06, 0.34]$); EUR/RON 0.31%
 - ▶ a price puzzle and an exchange-rate puzzle (depreciation after a tightening): the ROBOR innovation still contains the BNR's reaction to inflation and to pressure on the leu
- ▣ Euribor shock: Romanian IP -0.50% after 12 months (band $[-0.77, 0.07]$); HICP -0.10% ; ROBOR 0.16 pp
- ▣ Honest summary: the recursive VAR identifies the euro-area block credibly and the domestic policy shock poorly

How much of Romania comes from the euro area?

Variable	EA shocks, 12 m	EA shocks, 36 m	ROBOR shock, 12 m	own shock, 12 m
IP Romania	46%	37%	8%	41%
HICP Romania	44%	55%	3%	42%
ROBOR 3M	30%	39%	59%	—
EUR/RON	25%	23%	6%	58%

- Forecast error variance decomposition of the recursive VAR(2); EA shocks = the three euro-area shocks together
- Euro-area shocks explain 46% of Romanian IP and 55% of HICP at three years; the domestic rate shock explains less than a tenth of either
- The shares are as credible as the ordering; with a block-exogenous VAR they are a lower bound for external influence

ECB policy shocks and Romania: local projections



- Shock: the pure monetary policy shock of [Jarociński and Karadi \(2020\)](#) (authors' update, one s.d. = 3.2 bp); 3 lags of all variables and of the shock (lag augmentation), monthly dummies; origins 2005:8–2025:10

Interpreting the spillover projections

- ▣ Relevance first: the euro-area 1-year rate rises by 0.03 pp on impact (SE 0.01); first-stage $F = 4.8$ (impact) and 4.0 (one month)
 - ▶ monthly averages dilute an announcement-day surprise: the shock is a weak instrument for monthly euro-area rates
- ▣ Romanian HICP: -0.02% after 12 months (SE 0.15), 0.37% after 24 (SE 0.25); EUR/RON -0.31% after 6 months (SE 0.17)
- ▣ Honest conclusion: with a weak external shock and 20 years of data, the spillovers to Romanian prices are not determined; a weak-IV-robust set is wide (Seminar 3, C1)

Conclusions for Romania

- ▣ Robust findings
 - ▶ euro-area shocks account for a large share of Romanian output and price fluctuations
 - ▶ a euro-area tightening lowers Romanian IP within a year in the recursive VAR
- ▣ Fragile findings
 - ▶ the effect of a domestic ROBOR shock on prices and the leu: puzzles signal an identification failure, not a fact about Romania
 - ▶ the size of ECB spillovers to Romanian inflation: the instrument is weak at a monthly frequency
- ▣ Next steps (projects)
 - ▶ BNR policy-meeting surprises from ROBOR or FRA quotes around announcements, used as a proxy
 - ▶ sign restrictions (a tightening appreciates the leu and lowers prices) and a Bayesian VAR with shrinkage ([Chapter 5](#))

Recap: Romania and euro-area spillovers

- Put the large neighbour first: the euro-area block is credibly exogenous within the month
- A recursive ROBOR shock produces price and exchange-rate puzzles
- External ECB shocks are weak instruments at a monthly frequency: report the uncertainty, do not hide it

An open question

- How strongly does ECB monetary policy pass through to Romanian inflation, and has the pass-through changed since 2022?
 - ▶ formal: $H_0: \theta_{12} = 0$, the 12-month response of Romanian HICP to a 25 bp euro-area rate shock, tested by an Anderson–Rubin set that is valid with weak instruments
 - ▶ falsified by a bounded 90% set that excludes zero for a pre-registered shock, horizon and sample
- Why it matters: Romania plans to join the euro area; the BNR needs to know how much of its inflation is imported
 - ▶ literature to start from: Jarociński and Karadi (2020), Stock and Watson (2018), Montiel Olea, Stock and Watson (2021), Plagborg-Møller and Wolf (2021), Li, Plagborg-Møller and Wolf (2024)

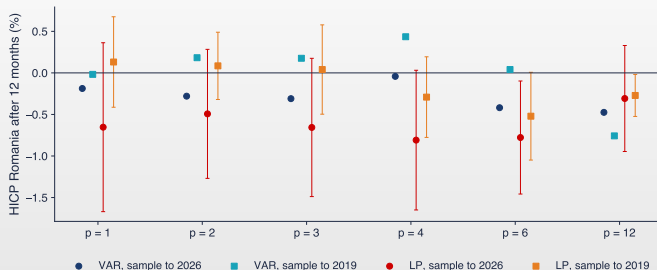
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
 - ▶ **literature:** List peer-reviewed papers that estimate euro-area monetary spillovers to Romania or Central and Eastern Europe with high-frequency shocks; give DOIs. Then check every DOI on Crossref
 - ▶ **hypothesis:** Propose three channels (trade, exchange rate, bank funding) and one observable implication of each.
 - ▶ **code and replication:** ask for an LP-IV function, then reproduce a known number first (the first-stage F of this lecture)
 - ▶ **robustness and critique:** Act as a hostile referee: list every way this identification of ECB spillovers could fail.
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- ▣ Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- ▣ The shock is dated correctly: announcement-day surprises are summed within the month, never shifted forward
- ▣ Relevance is reported (first-stage F with HAC variance) before any response is interpreted
- ▣ Lags, horizons, sample and the treatment of 2020 are fixed before the results; all variants are reported
- ▣ An AI summary of a “significant spillover” is checked against the confidence set, not against the point estimate

Mini-case: how robust is one number?



- Response of Romanian HICP after 12 months to a 25 bp Euribor shock: recursive VAR and recursive LP (90% Newey–West bars), lag length p , samples ending in 2019 and in 2026
- 24 estimates range from -0.81% to 0.44% , 7 of them positive: the sign depends on the lag length and on the post-2020 data; an AI assistant that reports one of them “with confidence” is wrong

Project idea

- ▣ **Euro-area monetary spillovers to Romania:** replicate first, then extend
 - ▶ replicate: the proxy SVAR of [Gertler and Karadi \(2015\)](#) (this lecture: $F = 17.7$, the bond-premium jump) and the Romanian LP of this lecture
 - ▶ extend: daily ECB surprises aggregated to monthly with the Gertler–Karadi averaging; a quarterly GDP version; Anderson–Rubin sets; sign restrictions on the surprises (JK)
 - ▶ pre-register: shock, outcomes, horizons 0–24, lag rule, the 2020 treatment and the Holm correction across outcomes
- ▣ Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- ▣ A structural VAR is a reduced form plus an identifying assumption; the assumption is the economics
- ▣ Zeros (short or long run), signs, narratives, heteroskedasticity and instruments each buy identification with a different assumption
- ▣ Set identification is honest: report the set and the role of the prior
- ▣ Inference: bootstrap matched to the setting, bias correction, joint bands, weak-IV-robust sets
- ▣ LP and VAR estimate the same responses; choose by bias and variance, and use LP for state dependence

Self-assessment

Questions

- How many restrictions identify a five-variable SVAR exactly?
- Why does $\mathbb{E}u_t z_t$ identify the impact column of a proxy SVAR?
- What does the posterior median of a sign-restricted response fail to represent?
- When does a VAR beat LP in mean squared error?
- Why is the wild bootstrap invalid for a proxy SVAR?

Next:

- Cointegration revisited: VECM, ARDL and panel data
- Long-run restrictions become restrictions on cointegrating vectors and on the common trends

Appendix: the long-run restriction in closed form

- ▣ Stable VAR: $\sum_{h \geq 0} \Phi_h = A(1)^{-1}$, so the long-run response matrix is $\Theta(1) = A(1)^{-1}B_0$
- ▣ $\Theta(1)\Theta(1)' = A(1)^{-1}B_0B_0'A(1)^{-1'} = A(1)^{-1}\Sigma_u A(1)^{-1'}$, known from the reduced form
- ▣ $[\Theta(1)]_{12} = 0$ makes $\Theta(1)$ lower triangular: $\Theta(1) = \text{chol}(A(1)^{-1}\Sigma_u A(1)^{-1'})$, unique with a positive diagonal
- ▣ Hence $B_0 = A(1)\Theta(1)$; check $B_0B_0' = \Sigma_u$; signs are then normalised (supply raises output in the long run)

Appendix: how the proxy identifies b_1

- $u_t = B_0 \varepsilon_t = b_1 \varepsilon_{1t} + \sum_{j \geq 2} b_j \varepsilon_{jt}$
- $\mathbb{E} u_t z_t = b_1 \mathbb{E} \varepsilon_{1t} z_t + \sum_{j \geq 2} b_j \mathbb{E} \varepsilon_{jt} z_t = \alpha b_1$ by exogeneity
- Relative impacts $b_{i1}/b_{11} = \mathbb{E} u_{it} z_t / \mathbb{E} u_{1t} z_t$ need $\alpha \neq 0$ (relevance); the scale of b_1 follows from $b_1' \Sigma_u^{-1} b_1 = 1$
- Equivalent 2SLS: regress u_{it} on u_{1t} with instrument z_t ; the first stage is the regression of u_{1t} on z_t

Appendix: LP and VAR agree up to horizon p

- ▣ Let $w_t = (y_{t-1}, \dots, y_{t-p})$ and the shock be the innovation $\tilde{x}_t = x_t - \text{proj}(x_t \mid w_t)$
- ▣ LP coefficient: $\beta_h = \text{Cov}(y_{t+h}, \tilde{x}_t) / \text{Var}(\tilde{x}_t)$ (Frisch–Waugh)
- ▣ The VAR(p) reproduces the autocovariances $\Gamma_0, \dots, \Gamma_p$ exactly; for $h \leq p$, $\text{Cov}(y_{t+h}, \tilde{x}_t)$ depends only on them, so both give the same β_h [Plagborg-Møller and Wolf \(2021\)](#)
- ▣ For $h > p$ the VAR extrapolates its own autocovariances: the source of the bias of a short VAR, and of the lower variance

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